

Kevin Chen

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EXPERIENCE

Amazon

Seattle, WA

Software Development Engineer II

July 2026 – Present

- Co-led launch of LLM-powered natural language search on **AWS Bedrock** (Anthropic Claude) for **30M MAU**, owning intent classification, query rewriting and suggestion generation pipelines with grounding against catalog metadata
- Owned quality and safety for the feature: **golden-set regression benchmark** with human relevance grading, **prompt-injection filtering**, and automatic fallback to lexical search on low-confidence or timed-out generations, achieving **~0.75 F1** on library-scoped title resolution, up from **~0.3** for the lexical baseline
- Designed and led the migration of Kindle Search to **AWS ECS**, introducing a **GraphQL gateway** and **asynchronous Kotlin-based service orchestration**, reducing infrastructure cost by **~20%**

Software Development Engineer

April 2024 – June 2026

- Owned development of the **Interactive Filter Refinement** and Android filter architecture, increasing filter usage by **540 TPS** and **monthly subscription borrows by 87K** (~\$794K annualized impact)
- Led the migration of Kindle Android Search UI to Jetpack Compose, **reducing UI lines of code by 54%** and **introducing reusable UI patterns adopted org-wide**
- Instrumented **200+ core service alarms and dashboards**; resolved **20+ accessibility** issues across **8 services**
- Mentored 3 new hires** and **won 2 internal hackathons** for LLM-based search personalization and content summarization

Software Development Engineer Intern

May 2022 – August 2022, May 2023 – July 2023

- Created an **AWS data pipeline** to ingest search metrics and analyze query trends using **DynamoDB, SQS, S3 and Lambda**
- Implemented the Kindle Search mobile UI using **ReactJS** for cross-platform UI modernization

University of Illinois Urbana-Champaign

Champaign, IL

Course Assistant, Computer Systems and Software Design Studio

December 2020 – May 2022

- Led discussion sessions and designed assignments on programming style and computer systems fundamentals

Computer Vision Research Assistant, Energy Transport Research Lab

May 2021 – December 2021

- Trained a Mask R-CNN model in PyTorch for liquid condensation detection, building the dataset with VGG Image Annotator

RESEARCH

Heterogeneous-Resource-Aware Federated Learning with Intelligent LoRA Allocation and Aggregation

2026 IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid) [ieeexplore.ieee.org/document/11619224]

Y. Xie, Y. Lian, **K. Chen**, A. Latif, L. Zhao, R. Farivar

- HRALoRA** adapts LoRA rank and aggregation to heterogeneous client memory and bandwidth, cutting communication cost **~50%** while improving accuracy on distributed fine-tuning benchmarks
- Contribution: designed and ran the evaluation across 2 base models and 6 data partitions spanning IID and non-IID splits, including ablations on rank allocation and bandwidth heterogeneity

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Master of Computer Science (GPA: 4.0/4.0)

May 2024 – August 2026

- Coursework:** *Deep Learning for Healthcare, Natural Language Processing, Applied Machine Learning, Cloud Computing*

Bachelor of Science, Computer Science + Linguistics (GPA: 4.0/4.0)

August 2020 – May 2023

- Activities:** President, LeetCode @ UIUC
- Awards:** UIUC Bronze Tablet Graduate (Top 3%), Dean's List, 2023 Outstanding Undergraduate Student

SKILLS

AI/ML: AWS Bedrock, PyTorch, NumPy, Pandas, OpenCV, HuggingFace PEFT, LoRA fine-tuning, RAG, vector search/embeddings, prompt engineering and evaluation, federated learning

Programming Languages: Kotlin, Python, Java, C++, TypeScript, SQL

Distributed/Backend: AWS (ECS, DynamoDB, SQS, CloudWatch, CloudFormation), GraphQL, Spring, Kotlin Coroutines

Frontend/Mobile: ReactJS, Jetpack Compose, Swift

Other Tools: Claude Code, Cline, Kiro, Docker, Git, Linux, Gradle, Maven, CI/CD, JUnit, Mockito, MockK

Spoken Languages: English (Native), Mandarin (Native), Japanese (JLPT N1), Spanish (Conversational)